MSG No. 90.102 P_I42_12 [Type IV, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `42'2'`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[1,2,3,6,12,13,15,16]`
2	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[4,5,7,8,9,10,11,14]`

Table 2: Wyckoff site: `2b`, site symmetry: `42'2'`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[1,2,3,6,12,13,15,16]`
2	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[4,5,7,8,9,10,11,14]`

Table 3: Wyckoff site: `4c`, site symmetry: `22'2'.`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[1,6,12,13]`
2	$\left[0, 0, \frac{3}{4}\right]$	`[2,3,15,16]`
3	$\left[0, 0, \frac{1}{4}\right]$	`[4,5,9,14]`
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[7,8,10,11]`

Table 4: Wyckoff site: `4d`, site symmetry: `2.22`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,6,7,8]`
2	$[0, 0, 0]$	`[2,3,4,5]`
3	$\left[0, 0, \frac{1}{2}\right]$	`[9,14,15,16]`
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[10,11,12,13]`

Table 5: Wyckoff site: `4e`, site symmetry: `4..`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, z\right]$	`[1,2,3,6]`
2	$\left[0, \frac{1}{2}, -z\right]$	`[4,5,7,8]`
3	$\left[0, \frac{1}{2}, z+\frac{1}{2}\right]$	`[9,10,11,14]`
4	$\left[\frac{1}{2}, 0, \frac{1}{2}-z\right]$	`[12,13,15,16]`

Table 6: Wyckoff site: `8f`, site symmetry: `2..`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[1,6]`
2	$[0, 0, z]$	`[2,3]`
3	$[0, 0, -z]$	`[4,5]`

continued ...

Table 6

No.	position	mapping
4	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	[7,8]
5	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	[9,14]
6	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[10,11]
7	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[12,13]
8	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	[15,16]

Table 7: Wyckoff site: 8g, site symmetry: ..2'

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[1,15]
2	$\left[-x,\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[2,13]
3	$\left[x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[3,12]
4	$\left[x+\frac{1}{2},\ -x,\ \frac{1}{4}\right]$	[4,11]
5	$\left[\frac{1}{2}-x,\ x,\ \frac{1}{4}\right]$	[5,10]
6	$\left[-x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[6,16]
7	$\left[x+\frac{1}{2},\ x,\ \frac{1}{4}\right]$	[7,9]
8	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{4}\right]$	[8,14]

Table 8: Wyckoff site: 8h, site symmetry: .2'.

No.	position	mapping
1	$\left[x,\ 0,\ \frac{3}{4}\right]$	[1,12]
2	$\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[2,15]
3	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[3,16]
4	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	[4,9]
5	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	[5,14]
6	$\left[-x,\ 0,\ \frac{3}{4}\right]$	[6,13]
7	$\left[0,\ x,\ \frac{1}{4}\right]$	[7,10]
8	$\left[0,\ -x,\ \frac{1}{4}\right]$	[8,11]

Table 9: Wyckoff site: 8i, site symmetry: .2'.

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	[1,12]
2	$\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[2,15]
3	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[3,16]
4	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	[4,9]
5	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{3}{4}\right]$	[5,14]
6	$\left[-x,\ 0,\ \frac{1}{4}\right]$	[6,13]
7	$\left[0,\ x,\ \frac{3}{4}\right]$	[7,10]
8	$\left[0,\ -x,\ \frac{3}{4}\right]$	[8,11]

Table 10: Wyckoff site: `8j`, site symmetry: `..2`

No.	position	mapping
1	$[x,\ x,\ 0]$	`[1,7]`
2	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ 0\right]$	`[2,5]`
3	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$	`[3,4]`
4	$[-x,\ -x,\ 0]$	`[6,8]`
5	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[9,15]`
6	$\left[-x,\ x,\ \frac{1}{2}\right]$	`[10,13]`
7	$\left[x,\ -x,\ \frac{1}{2}\right]$	`[11,12]`
8	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[14,16]`

Table 11: Wyckoff site: `16k`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ z\right]$	`[2]`
3	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ z\right]$	`[3]`
4	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	`[4]`
5	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ -z\right]$	`[5]`
6	$[-x,\ -y,\ z]$	`[6]`
7	$[y,\ x,\ -z]$	`[7]`
8	$[-y,\ -x,\ -z]$	`[8]`
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[9]`
10	$\left[-y,\ x,\ z+\frac{1}{2}\right]$	`[10]`
11	$\left[y,\ -x,\ z+\frac{1}{2}\right]$	`[11]`
12	$\left[x,\ -y,\ \frac{1}{2}-z\right]$	`[12]`
13	$\left[-x,\ y,\ \frac{1}{2}-z\right]$	`[13]`
14	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	`[14]`
15	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[15]`
16	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	`[16]`